

|                               |                               |                        |
|-------------------------------|-------------------------------|------------------------|
| <b>Engineering Physics II</b> | <b>Faculty of Engineering</b> | <b>Date: 3 /8/2025</b> |
| <b>B<sub>1</sub> MI/E</b>     | <b>First Test – Modeule C</b> |                        |
| <b>Name:</b>                  | <b>Reg. No.:</b>              |                        |

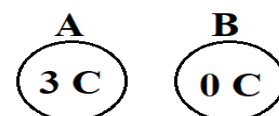
**Section I: Answer all the following by putting true (T) or false (F):** **6 Marks**

|                                                          |  |  |
|----------------------------------------------------------|--|--|
| 1) In Pascal principle, the force does change            |  |  |
| 2) A suck pressure is a negative gauge pressure          |  |  |
| 3) The electric flux does not change with electric field |  |  |

**Section II: Answer all the following by selecting the correct choice:** **14 Marks**

|                                                                                            |          |            |               |                  |
|--------------------------------------------------------------------------------------------|----------|------------|---------------|------------------|
| 1) Fully immersed object having a density more than the liquid's density. The object will: |          |            |               |                  |
| a) Move up                                                                                 | b) float | c) Sinking | d) be at rest | e) None of these |

2) If the two spheres shown below. The charge on sphere B is:

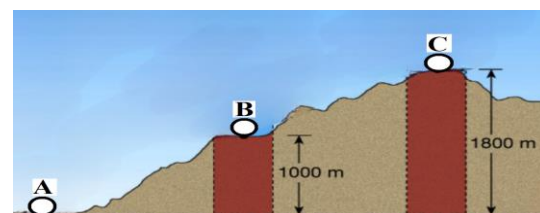


|          |        |          |        |                  |
|----------|--------|----------|--------|------------------|
| a) 1.5 C | b) 0 C | c) - 1 C | d) 3 C | e) None of these |
|----------|--------|----------|--------|------------------|

3) Which of the following has no charge:

|                      |            |           |             |                  |
|----------------------|------------|-----------|-------------|------------------|
| a) Electron & Proton | b) Neutron | c) Proton | d) Electron | e) None of these |
|----------------------|------------|-----------|-------------|------------------|

4) From the figure shown below. The pressures



|                |                |                |                |                  |
|----------------|----------------|----------------|----------------|------------------|
| a) $P_A > P_C$ | b) $P_B > P_A$ | c) $P_C > P_B$ | d) $P_C = P_A$ | e) None of these |
|----------------|----------------|----------------|----------------|------------------|

5) 25 N object in air and 10 N in water. The Buoyant force is:

|         |         |         |        |                  |
|---------|---------|---------|--------|------------------|
| a) 10 N | b) 25 N | c) 15 N | d) 0 N | e) None of these |
|---------|---------|---------|--------|------------------|

6) The force 8 N. If a distance changes from 8 m into 2 m. The force is:

|        |          |         |          |                  |
|--------|----------|---------|----------|------------------|
| a) 1 N | b) 0.5 N | c) 16 N | d) 128 N | e) None of these |
|--------|----------|---------|----------|------------------|

7) Object with  $4\text{ m} \times 3\text{ m} \times 2\text{ m}$  having a mass of  $x$  kg. The pressure in Pa on the ground is:

|          |            |         |             |                  |
|----------|------------|---------|-------------|------------------|
| a) $20x$ | b) $0.83x$ | c) $2x$ | d) $0.083x$ | e) None of these |
|----------|------------|---------|-------------|------------------|

**Section III: Answer all the following by selecting the correct choice**

**Full and clear steps are required**

**20 Marks**

| Constant                                |                                                              |                                          |
|-----------------------------------------|--------------------------------------------------------------|------------------------------------------|
| Electron: $m = 9.11 \times 10^{-31}$ kg | $K_e = 9 \times 10^9$ N.m <sup>2</sup> /C <sup>2</sup>       | Gravity: $g = 10$ m/s <sup>2</sup>       |
| Proton: $m = 1.67 \times 10^{-27}$ kg   | $G = 6.67 \times 10^{-11}$ N.m <sup>2</sup> /kg <sup>2</sup> | Water: $\rho_f = 1000$ kg/m <sup>3</sup> |

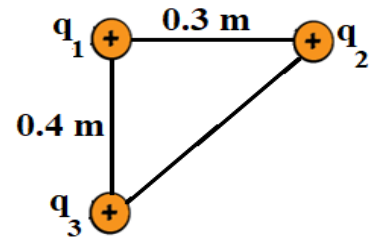
1) Two objects having  $q_1 = 2$  nC and  $q_2 = 2$  nC, separated by 30 cm. The electrostatic force is:

|                           |          |          |                          |                  |
|---------------------------|----------|----------|--------------------------|------------------|
| a) $400 \times 10^{-9}$ N | b) 0.4 N | c) 0.2 N | d) $60 \times 10^{-9}$ N | e) None of these |
|---------------------------|----------|----------|--------------------------|------------------|

2) 700 N object in air and 200 N when it is fully immersed in a fluid of  $\rho_f = 1500 \text{ kg/m}^3$ . The volume of the metal is:

|                       |                        |                       |                        |                  |
|-----------------------|------------------------|-----------------------|------------------------|------------------|
| a) $0.01 \text{ m}^3$ | b) $0.033 \text{ m}^3$ | c) $0.13 \text{ m}^3$ | d) $0.023 \text{ m}^3$ | e) None of these |
|-----------------------|------------------------|-----------------------|------------------------|------------------|

3) The total electric force on  $q_3$  is:  $q_1 = 1 \text{ } \mu\text{C}$ ,  $q_2 = 2 \text{ } \mu\text{C}$  and  $q_3 = 3 \text{ } \mu\text{C}$



|           |           |           |           |                  |
|-----------|-----------|-----------|-----------|------------------|
| a) 0.46 N | b) 0.75 N | c) 0.37 N | d) 0.57 N | e) None of these |
|-----------|-----------|-----------|-----------|------------------|

4) A room has dimensions of 4 m and 2 m and a height of 3 m. If the air density is  $1.21 \text{ kg/m}^3$ . The atmosphere force on the floor of the room is: Take  $P_{\text{atm}} = 10^5 \text{ Pa}$

|             |             |             |           |                  |
|-------------|-------------|-------------|-----------|------------------|
| a) 900000 N | b) 600000 N | c) 800000 N | d) 500000 | e) None of these |
|-------------|-------------|-------------|-----------|------------------|

5) Two opposite charges one is 4 times the other separated by distance  $r$ , if a ratio of charge to a distance  $r$  is  $2 \text{ } \mu\text{C/m}$ . The electric force on  $3q$  placed at distance  $r$  from a point where  $E = 0$  is:

|          |           |        |           |                  |
|----------|-----------|--------|-----------|------------------|
| a) 1.2 N | b) 13.3 N | c) 0 N | d) 1.33 N | e) None of these |
|----------|-----------|--------|-----------|------------------|